2e

SSLC KARNATAKA

Mathematics

March- 2018

Class-X Time: 3 Hours **Max. Marks: 80**

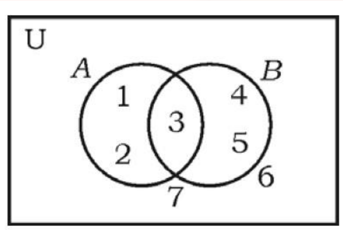
General Instructions to the Candidate:

(i) This Question Paper consists of 40 objective and subjective of questions.
(ii) This questions paper has been sealed by reverse jacket. You have to cut on the right side to open the paper at the time of commencement of the examination .Check whether all the pages of the question paper are intact.
(iii) Follow the instructions given against both the objective and subjective types of questions.
(iv) Figures in the right hand margin indicate maximum marks for the questions.
(v) The maximum time to answer the paper is given at the top the question paper. It includes 15 for minutes for reading the question paper.

SECTION - A

I. Four alternatives are given for each of the following questions /incomplete statements. Only one of is correct or most appropriate. Choose the correct alterative and writhe the complete answer along with its letter of alphabet.

Q 1. In the given Venn diagram $n(A)$ is **[8 x1 =8]**



(A) 3 (B) 1` (C) 5 (D) 2

Ans 1 (A) 3

Q 2 Sum of all the first 'n' terms of even natural number is

(A) $n(n+1)$ (B) $n(n+2)$ (C) n^2 (D) $2n^2$

Ans 2. (A) $n(n+1)$

Q 3. A boy has 3 shirts and 2coast. How many different pairs, a shirt and a cost can he dress up with?

(A) 3 (B) 18 (C) 6 (D) 5

Ans 3. (C) 6

Q 4. In a random experiment, it the occurrence of one event prevents the occurrence of other event, it is

(A) A complementary event (B) a certain event

(C) Not mutually exclusive event (D) mutually exclusive event

Ans. 4. (D) mutually exclusive event

Q 5. If the polynomial $p(x)=x^2-x+1$ is divided by $(x-2)$ than the remainder is

(A) 2 (B) 3 (C) 0 (D) 1

Ans 5. (B) 3

Q 6. The distance between the co-ordinates of a point (p,q) form the origin is

(A) p^2-q^2 (B) $\sqrt{p^2-q^2}$ (C) $\sqrt{p^2+q^2}$ (D) q^2-p^2

Ans 6. (C) $\sqrt{p^2+q^2}$

Q 7. The equation of line having slope 3 and y-intercept 5 is

(A) $3y=5x+3$ (B) $5y=3x+5$

(C) $y=3x-5$ (D) $y=3x+5$

Ans 7. (D) $y=3x+5$

Q 8. The surface area of a sphere of radius 7 cm is

(A) 88 cm^2 (B) 616 cm^2 (C) 661 cm^2 (D) 308 cm^2

Ans 8. (B) 616 cm^2

SECTION -B

II. Answer the following: [6 x 1 =6]

9. Find the HCF of 14 and 21.

Ans 9.) 14 21

Λ Λ

2 7 3 7

HCF =7

Q 10. The average run scored by a batsman in 15 cricket matches is 60 and standard deviation of the runs is 15. Find the coefficient of variation of the runs scored by him.

Ans 10. × =60 S. D = 15

$$\text{C.V} = \frac{\sigma}{\times} = 100 = \frac{\cancel{15}^{5}}{\cancel{60}} \times \cancel{1}00$$

C.V = 25

Q 11. Write the degree of the polynomial $f(x) = x^2 - 3x^3 + 2$

Ans 11. Degree = 3

Q 12. What are congruent circles?

Ans 12. The circles having same radius but different are called congruent circles.

Q 13. If $\sin\theta = \frac{5}{13}$ than write the value of cosec θ

Ans 13. $\sin\theta = \frac{5}{13}$

$$\text{Cosec} = \frac{1}{\sin\theta} = \frac{1}{5/13}$$

$$\text{Cosec} = \frac{13}{5}$$

Q 14. Write the formula used to find the total surface area of a right circular cylinder.

Ans 14. Total Surface Area = $2\pi r\left(r+h\right)$ square units .

SECTION -C

Q 15. If U ={0,1,2,3,4} and A = {1,4}, B ={1,3} show that (A$\cup$ B)^{I} =A^{I} $\cap$ B^{I} . 2

Ans 15. Given if {u =20 ,1,2,3,4}and A ={1.4} B ={1,3}

(AUB) =A^1 $\cap$ **B**^{1}

(AUB) =U-AUB

={ 2,0,1,2,3,4,} –[{1,4,} U{1,3}]

={0,$\cancel{1}$ 2,$\cancel{3}$,$\cancel{4}$ } {$\cancel{1}$,$\cancel{3}$,$\cancel{4}$}

(AUB) ={0,2} →1

A^1 $\cap$ **B**^{1} **=U -** A^1 $\cap$ U-B

= [{0, $\cancel{1}$ 2,$\cancel{3}$,$\cancel{4}$} –{$\cancel{1}$,$\cancel{4}$ }] $\cap$[{ 0, ,$\cancel{1}$ 2,$\cancel{3}$,$\cancel{4}$} –{$\cancel{1}$,$\cancel{3}$ }]

= { 0,2,3,} $\cap$ {0,2,4}

={ 0, 2} →2

=1=2

(A$\cup$ B)^1 =(A$\cap$B)^1

Q 16. Find the sum of the series 3+7+11+…………..to 10 terms. 2

Ans 16. A=3 ,d= j2 1 1=j=3

$$\frac{1\cancel{0}^{5}}{2}[2(a)+(a-1)-4]$$

$$3_{10}== 5(6+36)$$

$$= 5(42)$$

$$5_{10} = 21$$

Q 17. At constant pressure certain quantity of water at 24°C is heated . It was observed that rise of temperature was found to be 4°C per minute . Calculate the time required to rise to the temperature of water to 100°C at sea level by using formula. 2

Ans 17). 24° ,28 ,32….100

a = 24, d= j-j= 28-24=4

j =100

j_A =(a +b-1) d

100 =24+(n-1) 4

100= 24 +4 =∩ -4

= 4 ∩ = 100-20

40 =80 =20

∩ =20

20 mites is required to rise the temperature to 100° C.

Q 18, Prove that 2+$\sqrt{5}$ is an irrational number. 2

Ans. 18). Let us assume 2+$\sqrt{5}$ is a rational 10

$$2+\sqrt{5} = \frac{p}{q},\ q \neq 0$$

$$\sqrt{5} = \frac{p}{q} = 2$$

$$\sqrt{5} = \frac{p = 2q}{q}$$

$\frac{p = 2q}{q}$ is a cantonal no but $\sqrt{5}$ is a irrational our assumption was wrong $2+\sqrt{5}$ is a an irrational as.

Q 19. If $^{n}p_{4} = 20(^{n}p_{2})$ then find the value of n . **2**

Ans 19). $^{n}p_{4} = 20(^{n}p_{2})$

$$\frac{a1}{(a-4)} = 20\left(\frac{a1}{}\right)$$

$$\frac{1}{(n-4)} = 20\left(\frac{1}{(n-2)(a-2)(a-1)}\right)$$

$$1 = \frac{20}{(n-2)(a-3)}$$

$$(a-2)(a-3) = 20$$

$$n^{2}3n - 2a + b = 20$$

$$a^{2-}5a + 6 - 2a = 0$$

$$n^{2} - 5a - 14 = 0$$

$$n^{2} - 1a + 2a - 14 = 0$$

$$n(a = 7) + 2(a - 7) = 0$$

$$(a-7)(a+2) = 0$$

$$a = 7n$$

$$\bigcap \neq 2 / acannotbenegetive$$

$$a = 7$$

Q 20. A dice numbered 1 to 6 on its faces is rolled once. Find the probability of getting either an even number or multiple of '3' on its top face. **2**

Ans. 20) S = {1, 2, 3, 4, 5, 6}

$\cap(s) = 6$

Lat A bean event of getting an even no

A = {2, 4, 6} a (A) =3

P(A) $= \frac{n(A)}{n(S)} = \frac{2}{6} = \frac{1}{2}$

Let B be an event of getting an multiple of s

{B =2, 3, 6} a (B) =2

P (B) = $\frac{a(B)}{a(S)} = \frac{2}{6} = \frac{1}{3}$

Event of getting either event

P (A$\cup$ B) =P (A) +P (B)

$$\frac{1}{2}+\frac{1}{3}$$
$$=\frac{3+2}{6}$$

p/a=UB=$\frac{5}{6}$

Q 21.What are the surds and unlike surds?

Ans. 21) A group of surds having order eradicate same area called like surds.

A group of surds hiving with or one order or vesicant different are celled us surds. Like surds ex:- $2\sqrt{2}$, $\sqrt{2}$,8$\sqrt{200}$, unlike surds ex:- $2\sqrt{3}$,$8\sqrt{8}$,8 : $3\sqrt{18}$ etc.

Q 22. Rationalise the denominator and simplify:

$$\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$$

Ans 22). $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$

R.F=conjugate = $\sqrt{5+\sqrt{3}}$

$$\frac{\left(\sqrt{5}+\sqrt{3}\right)\left(\sqrt{5}+\sqrt{3}\right)}{\left(\sqrt{5}+\sqrt{3}\right)\left(\sqrt{5}-\sqrt{3}\right)}$$

$$\left(\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5^1-\sqrt{3^1}}}\right)^2$$ /(a+b) (a-b)=$a^{2-}b^2$

$= \dfrac{\sqrt{5} + B + 2\sqrt{5 - \sqrt{3}}}{5 - 3}$

$= \dfrac{5 + 3 + 2\sqrt{5}}{2}$

$= \dfrac{8 + 2\sqrt{5}}{2}$

$= \dfrac{\cancel{2}(4 + \sqrt{5}}{\cancel{2}}$

$= 4 + \sqrt{15}$

Q 23. Find the quotient and the remainder when $f(x) = 2x^3 - 3x^2 5x - 7$ is divided by $g(x) = (x - 3)$ using synthetic division. **2**

OR

Find the zeros of the polynomial $p(x) = x^2 - 15x + 50$.

$(x) = 2x^3 - 3x^2 + 5x - 7$

Ans 23). f $g(x) = x - 3$

$x - 3 = 0, x = 3$

3 2 -3 5 -7

$\downarrow$ 6 9. 42

2 3 14 /35/ $\rightarrow$ r(x)

$q(x) = 2x^2 + 3x + 14$

$r(x) - 35$

Q 24. Solve the equation $x^2 - 12x + 27 = 0$ by using formula.

Ans 24). $x^2 - 12x + 27 = 0$

a = 1, b = 12, c =27

$x = -b + \dfrac{\sqrt{b^2} - 4_a}{2_a}$

$$= \frac{(12) + \sqrt{(12)^2 - 4 \times 1 \times 27}}{2 \times 1}$$

$$\frac{12 + \sqrt{144 = 108}}{2}$$

$$= \frac{12 + \sqrt{36}}{2}$$

$$= \frac{12 + 6}{2}$$

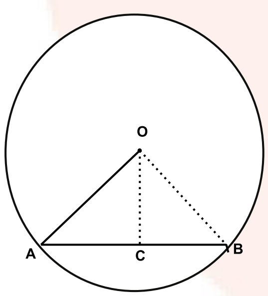
$$= \frac{x = 12 + 6}{2} \quad \text{or} \quad \frac{x = 12 - 6}{2}$$

$$= \quad x = 18/_2 \quad \text{0r} \quad x = 36/_2$$

$$x - 9 \quad \text{or} \quad x - 3$$

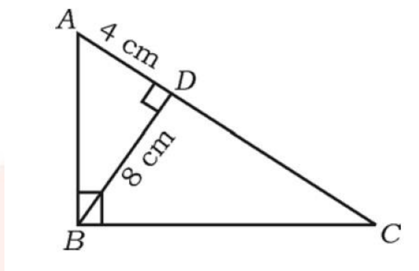
Q 25. Draw a chord of length 6 cm in circle of radius 5 cm. Measure and write the distance of the chord form the center of the circle. 2

Ans 25.



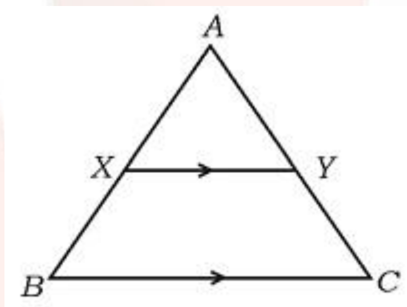
The distance of the chord form the center of the circle is 4cm.

Q 26. In A ABC, ∠ABC =90°, BD ⊥ AC.If BD =8 cm ,AD =4 cm , find CD and AB. 2



OR

In Δ ABC, XY || BC and XY= $\frac{1}{2}$ BC. If the area of Δ AXY =10 cm^2 ,find the area of trapezium XYCB.



Ans 26. In Δ ABC

Given ABC 10 BD AC

CD =9

In Δ ABC, Δ AD

∠B=∠D =90

∠c = ∠c

Δ ABC~Δ BDC

BD^2 =AD D1

8^2 = 4 × 1

16 $\cancel{64}$ = $\cancel{4}$ CD

$AB^2 = 80 = AB = \sqrt{80}$

$AB = 4\sqrt{5}cm$

Q 27. Show that, cost θ + sin θ = cosec θ **2**

Ans 27). Cost θ + sin θ = cosec θ

$= \frac{cosec\theta}{sin\theta} = cosec\theta + sin\theta$ | $cost\theta \frac{cos\theta}{sin\theta}$

$\frac{\mathbf{cos}^2\theta}{\mathbf{sin}\,\theta}$

$= = \frac{\mathbf{cos}^2\theta + \mathbf{sin}^2\theta}{\mathbf{sin}\,\theta}$

$= \frac{1}{\mathbf{sin}\,\theta}$

$cose1\theta$

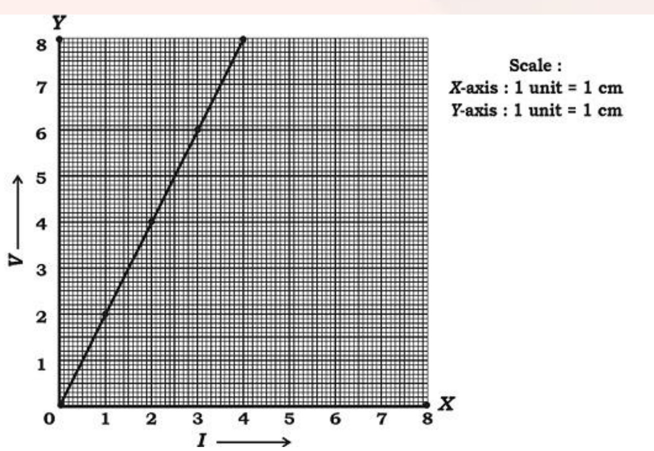
Q 28. A student while conducting an experiment on Ohm's law, plotted the graph according to the given data. Fine the slop of line obtained. **2**

X- axis I	1	2	3	4
Y-axis V	2	4	6	8

Scale:

X –axis : 1 unit =1cm

X –axis: 1 unit = 1 c



$(x, y,) = (1, 2)$

$(x_2 + y_2) = (2, 4)$

$(x_3 + y_3) = (3, 4)$

Ans 28. $slope = \dfrac{y_4 = y_4}{x_4 = x_4} = \dfrac{8,-2}{4-1}$

$\dfrac{62}{6}$

$slope = 2$

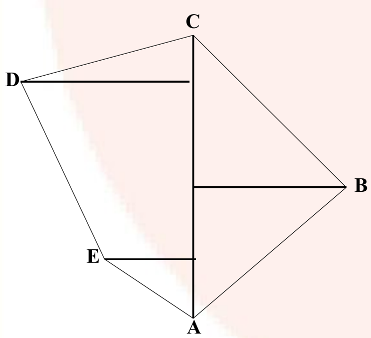
Q 29. Draw the plan for the information given below: 2

Scale 20 m =1 cm

	Metre To C	
To D 50	140	
	100	40 to B
	60	
To E 30	40	
	From A	

Ans. 29.Scale 20 m =1 cm

Ans:



Q 30. Out of different bicycle companies, a student likes to choose bicycle form three companies .Find out in out in how many ways he can choose the companies to buy bicycle. 2

Ans. 30)

$$8i_3 = \frac{8\,!\;\;3}{(8-3)!\;\;3\,!}$$

$$\frac{8\times7\times6\times5}{5!\times3\times2}$$

$$=56$$

In 56 ways he can choose & different companies out of 8.

SECTION -D

Q 31.) In a Geometric progression the sum of fist three terms is 14 and the sum of next three terms of it is 112.Find the Geometric progression. 3

OR

If'a' is the Arithmetic mean of b and c,'b' is the Geometric mean of c and a ,then prove that 'e' is the Harmonic mean of a and b.

Ans. 31) Given

$$j_1 + j_2 + j_3 = 14$$

$$J_4 + j_5 + j_6 = 112$$

$$a + ar + ar^2 = 14$$

$$a(1+r+r^2) = 14 \rightarrow 1$$

$$ar^3 + ar^4 + ar^5 = 112$$

$$ar^3\left(1+r+r^2\right) = 112\,2$$

$$2 \div 1$$

$$\cancel{a}r^3(14+r+r^2) = 11\cancel{0}$$

$$\cancel{a}(1+r+r) = 1\cancel{4}$$

Or

$r^3 = 8$

$r^3 = 2^3$

$r = 2$

$a(1+r+r^2) = 14$

$a(1+2+4) = 14$

$a = \frac{14}{7}$

$a = 2$

$a = 2, ar = 2\times 2 = 4, ar^2 2\times 2^2 = 8$

$G.P = 2, 4, 8, 16.$

Q 32. Marks scored by 30 students of 10th standard in a unit test of mathematics are given below. Find the variance of the scored. 3

Marks (x)	4	8	10	12	16
No. of students (f)	13	6	4	3	4

Ans. 32)

x	f	fx	D=x –x	D^2	Fd^2
4	15	52	4-8=4	16	208
8	6	48	8-8=0	0	0
10	4	40	10-8=2	4	16
12	3	36	12-8=4	16	48
16	4	66	16-8=8	64	256
	∩=30	Fx=240			$E,fd^2=528$

$$\frac{fx}{\cap} = \frac{\cancel{24}^{8}\cancel{0}}{3\cancel{0}}$$

X= ε^{x-8}

$$V = \frac{\varepsilon fd^2}{\cap} = V \approx 17.6$$

Q 33 If p and q are the roots of the equation $x^2 - 3x + 2 = 0$**,** $find the value of \frac{1}{q} - \frac{1}{q}$ **3**

OR

A dealer sells an article for Rs. 16 and loses as much per cent as the cost price of article. Find the cost of the article.

Ans 33). It travelled 3am in 24 sec velocity= 300mls

$$x^2 + 3, x + 2 = 0$$
$$= a \le 1, 20b = 3, c - 2$$
$$sum = p + q = b / a = c - 3$$
$$p + q = 2 - 3$$
$$prodect = pq = y_a 2 / 1 = 2$$
$$pq = 2$$
$$\frac{q = q - e}{p = q = p_q}$$
$$= \overline{)2 + q = 4aq}$$
$$= \overline{)3^2 - 4 \times 2}$$
$$= 9\overline{)9 - 8}^{\,2}$$
$$= \frac{\sqrt{1}}{2}$$
$$\frac{1}{p} = \frac{1}{q} = 1 / 2$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$
$$(a+b)^2 - 2ab^2 a^2 + b^2 \rightarrow$$
$$(a2b\_)^2 = a^2 + b^2 + = 1$$
$$form 1 = 2Raform$$
$$(a+b)^2 = (a+b)^2 = a^2 b = 2ab$$
$$(a+b)^2 = (a+b)^2 2ba = bab$$
$$(a-b) = \overline{)(a+b)}^2 - ab$$
$$\frac{1}{6}$$

Q 34. Prove that "If two circles touch other externally, their centers and the point of contact are collinear." 3

Ans 34). Data A is the center of c_1

B 1s the center of c_2

P is the point of contact

To prove =A, B, ϵ pare collinear

Construction –Draw tangent x

Prof statement Reason

∟APX=90→1 {Radius is perpendicular

∟BPx =90→2 { to the tangent dram.

1+2

∟APx + ∟BPx =90° +90° =180

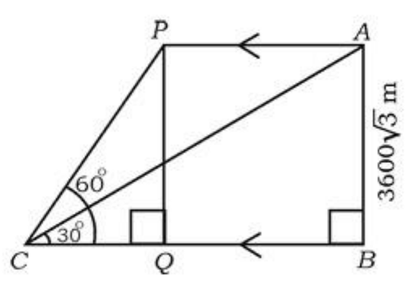
180° is angle formed is a straight line

A,B,ε P are collinear.

Q 35. If $\sin^2\theta + 3\cos^2\theta = 4$ and 'θ' is acute then show that $\cot\theta = \sqrt{3}$. (3)

OR

The angle of elevation of an aircraft from a point on horizontal ground is be 30°. The angle of elevation of same aircraft 24 seconds which is moving horizontal to the ground is found to be 60°. If the height of the aircraft from the ground is $3600\sqrt{3}$ meters, find the velocity of the aircraft.



. 35) In Ans Δ ABC, Jan 30°= $\frac{AB}{BL}$ given AB=3600 $\sqrt{3m}$

$$\frac{1}{\sqrt{3}} = \frac{3600}{BC} \int AB3600\sqrt{3}$$

$Bc3600\sqrt{3}$ ∟ACB =30°

$Bc10800m$

∟PCB =60°

In Δ CPQ=-

Jan ε $= \frac{PQ}{QC}$

In= $\frac{3600\sqrt{3}}{QC}$

QC=3600 m

BQ =BC-QC

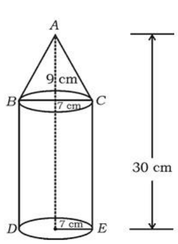
=10800-3600

BQ=7200 m

BQ =PA=7200 m

&= $\frac{d}{t}$ =7200 m it travelled 300 m

Q 36. A solid is in the form of a cone mounted on a right circular cylinder, both having same radii as shown in the figure. The radius of the base and height of the cone are 7 cm and 9 cm respectively. If the total height of the solid is 30 cm, find the volume of the solid. 3



OR

The slant height of the frustum of a come is 4 cm and perimeter of its circular bases are 18 cm and 6 cm respectively. Find the curved surface area of the frustum.

Solution:

We know that

Curved surface area of frustum = $\pi\left(r_1+r_2\right)l$

Here, $l=4cm$

We need to find r_1 and r_2

For r_1

Circumference of 1st end = 18 cm

$$2\pi r_1=18$$
$$r_1=\frac{18}{2}\times\frac{1}{\pi}$$
$$r_1=\frac{9}{\pi}$$

For r_2

Circumference of 1st end = 18 cm

$$2\pi r_2=6$$
$$r_2=\frac{6}{2}\times\frac{1}{\pi}$$
$$r_2=\frac{3}{\pi}$$

SECTION-E

Q 37. Solve the equation $x^2-x-2=0$ graphically. **4**

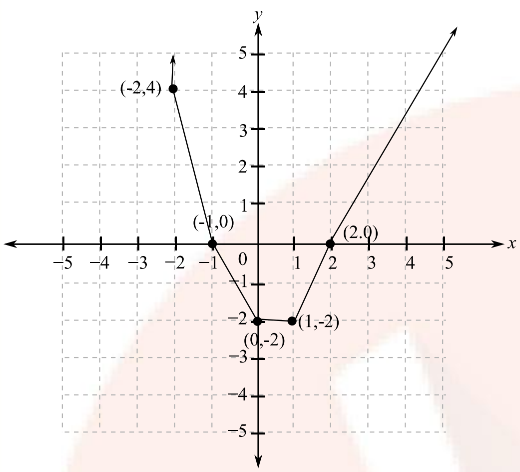
Ans:

$$y=x^2-x-2$$

Now for different values of x, we will find the values of y

x	1	2	0	-1	-2
y	-2	0	-2	0	4

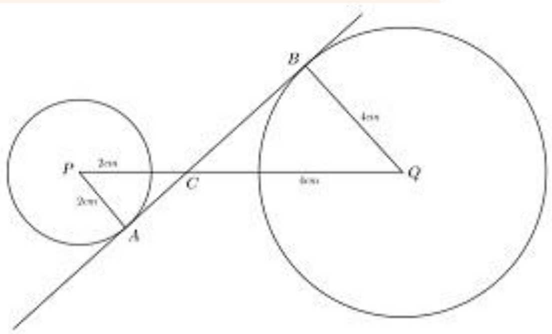
Now we will plot it on graph as below:



Q 38. Construct a direct common tangent to two circles of radii 4 cm and 2 cm whose centres are 9 cm apart, Measure and write the length of the tangent. 4

Ans 38) given R =4 cm

=2 cm



Draw a line segment PQ of length 9cm.

With P as a centre draw a circle of radius 2cm and with Q as centre draw a circle with radius 4cm.

Let A and B be the points where the transverse tangent touches these two circles.

Let C be the intersection point of AB and PQ.

In ΔAPC and ΔBQC

$\angle PAC = \angle QBC = 90°$ (Tangent radius property)

$\angle PCA = \angle QCB$ (Vertically opposite angles)

$\therefore \Delta APC \sim \Delta BQC$ (AA test)

$\Rightarrow \frac{PC}{QC} = \frac{PA}{QB} = \frac{2}{4} = \frac{1}{2}$

$\Rightarrow 2PC = QC$

Given that: $PC + QC = 9$

$$PC + QC = 9$$
$$PC + 2PC = 9$$
$$3PC = 9$$
$$PC = 3$$
$$QC = 6$$

$$AC = \sqrt{PA^2 + PC^2}$$
$$AC = \sqrt{4+9}$$
$$AC = \sqrt{13}$$

Similarly,

$$BC = \sqrt{BQ^2 + QC^2}$$
$$BC = \sqrt{16+36}$$
$$BC = \sqrt{52}$$

Length of AB = AC + BC = $\sqrt{13} + \sqrt{52} = \sqrt{65} = 8.06cm$

Q 39. State and prove Basic Proportionality (Thale's) theorem.

Basic Proportionality Theorem states that, if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides of the triangle in proportion.

In the given figure, if we consider DE is parallel to BC, then according to the theorem,

AD/BD=AE/CE

Proof:

$Area\ of\ Triangle\ =\ ½\ \times\ base \times\ height$

In ΔADE and ΔBDE,

$$\frac{Ar\left(ADE\right)}{Ar\left(DBE\right)}=\frac{\frac{1}{2}\times AD\times EF}{\frac{1}{2}\times DB\times EF}=\frac{AD}{DB}\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

In ΔADE and ΔCDE,

$$\frac{Ar\left(ADE\right)}{Ar\left(ECD\right)}=\frac{\frac{1}{2}\times AE\times DG}{\frac{1}{2}\times EC\times DG}=\frac{AE}{EC}\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

ΔDBE and ΔECD have a common base DE and lie between the same parallels DE and BC. The triangles having the same base and lying between the same parallels are equal in area.

So, we can say that

$$Ar\left(\Delta DBE\right)=Ar\left(\Delta ECD\right)$$

Therefore,

$$\frac{A\left(\Delta ADE\right)}{A\left(\Delta BDE\right)}=\frac{A\left(\Delta ADE\right)}{A\left(\Delta CDE\right)}$$

Therefore, from (1) and (2)

$$\frac{AD}{BD}=\frac{AE}{CE}$$

Hence Proved.

Q 40. A vertical tree is broken by the wind at height of 6 meter from its foot and top touches the ground at a distance of 8 meter from the foot of the tree. Calculate the distance between the top of tree before breaking and the point at which tip of the touches the ground, after it breaks. 4

OR

In Δ ABC, AD is drawn perpendicular to BC: CD=3:1, than prove $BC^2=2(AB^2-AC^2)$.

Ans 40). In Δ CBD, ∟B=90°

Given -∟B =6m

BD = 8m

AD= 2

In Δ CBD –B=90° the distance b/w the top of the

$CD^2=BC^2+BD^2$ tree before breaking ε the point

$CO^2=6^2+8^2$ at which tip of the tree touches

$CD^2 =36+64$ the ground after it breakers is

CO = $\sqrt{100}$

CO=10 m

CD=AC [broken part]

C0=10 m AC

In Δ ABC, 10=90°

$AD^2 \quad AB^2 + BD^2$

$AD^2= (AC+CB)^2 +8^2$

$AD^{2=} (6+10)^{2+}64$

$AD^{2=}16^2 +64$

AD^2 256+64

AD= $\overline{)320 \approx 17.8m}$

17.8 m

OR

Ans40) Given BD: 10 =3:1

BD & CD

To prove BC^2 $2(AB^2 - AC^2$

In Δ ADC, $\llcorner D = 90°$

1 $\leftarrow$ $AC^2 = AO^2 + CO^2$ | from Pythagoras tertian.

In Δ ABD, $\llcorner \underline{\underline{D}}$ 90°

From Pythagoras tertian.

$AB^2 = AD^2 + BD^2$ 2

Consider RHS

$= 2(AB^{2-}AC^2)$

$= 2[(AD^2+BD^2)\ (AD^2+CO^2)]$

$= 2(AD^2+BD^2)\ (AD^2+CO^2)$

$=2(BD^2-CO^2)$

$=2\ [\left(\frac{3BC}{4}\right)^2 = \left(\frac{1BC}{4}\right)^2]$

$=2\left(\left(\frac{3BC}{4}\right)^2 = \left(\frac{BC}{4}\right)^2\right)$

$= 2\left(\frac{9BC^2}{16} - \frac{BC^2}{16}\right)$